

Day2-Case Study: Hospital Management System

↑ patients							
	_id ObjectId	patientName String	doctor String	department String	age Int32	treatmentCost Int32	daysAdmitted
1	ObjectId('6a115d2dc043d7...	"Arun"	"Dr. Ravi"	"Cardiology"	45	20000	5
2	ObjectId('6a115d2dc043d7...	"Meena"	"Dr. Kumar"	"Neurology"	50	30000	7
3	ObjectId('6a115d2dc043d7...	"Raj"	"Dr. Ravi"	"Cardiology"	60	15000	3
4	ObjectId('6a115d2dc043d7...	"Divya"	"Dr. Mehta"	"Orthopedics"	35	10000	2
5	ObjectId('6a115d2dc043d7...	"Kiran"	"Dr. Kumar"	"Neurology"	40	25000	6
6	ObjectId('6a115d2dc043d7...	"Sneha"	"Dr. Mehta"	"Orthopedics"	30	8000	2
7	ObjectId('6a115d2dc043d7...	"Vijay"	"Dr. Ravi"	"Cardiology"	55	22000	4
8	ObjectId('6a115d2dc043d7...	"Priya"	"Dr. Kumar"	"Neurology"	48	27000	5
9	ObjectId('6a115d2dc043d7...	"Anil"	"Dr. Mehta"	"Orthopedics"	52	12000	3
10	ObjectId('6a115d2dc043d7...	"Riya"	"Dr. Ravi"	"Cardiology"	29	18000	4

1. Find all patients whose treatment cost is greater than 20,000 and display only their name and department.

```
> db.patients.aggregate([
  {$match:{treatmentCost:{$gt:20000}}},
  {$project:{patientName:1,department:1,_id:0}}
])
```

2. List all patients sorted by treatment cost in descending order.

```
> db.patients.aggregate([
  {$sort:{treatmentCost:-1}}
])
```

3. Display the first 5 patient records.

```
> db.patients.aggregate([
  {$limit:5}
])
```

4. Skip first 3 patients and display the remaining records.

```
> db.patients.aggregate([
  {$skip:3}
])
```

5. Find patients from the Cardiology department and display only their names.

```
> db.patients.aggregate([
  {$match:{department:"Cardiology"}},
  {$project:{patientName:1,_id:0}}
])
```

6. Find total treatment cost per department.

```
> db.patients.aggregate([
  {$group:{_id:"$department",TotalTreatmentCost:{$sum:"$treatmentCost"}}},
  {$project:{department:"$_id",TotalTreatmentCost:1,_id:0}}
])
```

7. Find average treatment cost per doctor and display results in ascending order.

```
> db.patients.aggregate([
  {$group:{_id:"$doctor",AverageTreatmentCost:{$avg:"$treatmentCost"}}},
  {$project:{doctor:"$_id",AverageTreatmentCost:1,_id:0}},
  {$sort:{AverageTreatmentCost:1}}
])
```

8. Find departments where total treatment cost is greater than 50,000.

```
> db.patients.aggregate([
  {$group:{_id:"$department",TotalTreatmentCost:{$sum:"$treatmentCost"}}},
  {$match:{TotalTreatmentCost:{$gt:50000}}},
  {$project:{department:"$_id",TotalTreatmentCost:1,_id:0}}
])
```

9. Find number of patients in each department and sort the result in descending order.

```
> db.patients.aggregate([
  {$group: {_id: "$department", PatientCount: {$sum: 1}}},
  {$project: {department: "$_id", PatientCount: 1, _id: 0}},
  {$sort: {PatientCount: -1}}
])
```

10. Find top 3 doctors with highest total treatment cost.

```
> db.patients.aggregate([
  {$group: {_id: "$doctor", TotalTreatmentCost: {$sum: "$treatmentCost"}}},
  {$project: {doctor: "$_id", TotalTreatmentCost: 1, _id: 0}},
  {$sort: {TotalTreatmentCost: -1}},
  {$limit: 3}
])
```

11. Find patients whose age is greater than 40, group them by department, and count number of patients.

```
> db.patients.aggregate([
  {$match: {age: {$gt: 40}}},
  {$group: {_id: "$department", NumberOfPatients: {$sum: 1}}},
  {$project: {department: "$_id", NumberOfPatients: 1, _id: 0}}
])
```

12. Find average treatment cost per department, display only department and average cost, and sort descending.

```
> db.patients.aggregate([
  {$group: {_id: "$department", AverageTreatmentCost: {$avg: "$treatmentCost"}}},
  {$project: {department: "$_id", AverageTreatmentCost: 1, _id: 0}},
  {$sort: {AverageTreatmentCost: -1}}
])
```

13. Find departments having more than 2 patients admitted.

```
> db.patients.aggregate([
  {$group: {_id: "$department", NumberOfPatients: {$sum: 1}}},
  {$match: {NumberOfPatients: {$gt: 2}}},
  {$project: {department: "$_id", _id: 0}}
])
```

14. Display top 3 departments based on highest total treatment cost after skipping the first one.

```
> db.patients.aggregate([
  {$group: {_id: "$department", TotalTreatmentCost: {$sum: "$treatmentCost"}}},
  {$project: {department: "$_id", TotalTreatmentCost: 1, _id: 0}},
  {$sort: {TotalTreatmentCost: -1}},
  {$skip: 1},
  {$limit: 3}
])
```

15. Find doctors treating patients with cost greater than 10,000, calculate total cost, and show only top 2 doctors.

```
> db.patients.aggregate([
  {$match: {treatmentCost: {$gt: 10000}}},
  {$group: {_id: "$doctor", Cost: {$sum: "$treatmentCost"}}},
  {$project: {doctor: "$_id", Cost: 1, _id: 0}},
  {$sort: {Cost: -1}},
  {$limit: 2}
])
```